

De Gouden Tarwe SQL Database Challenge

(Day 1)

MySQL

Problem Statement

De Gouden Tarwe (Dutch for The Golden Wheat) is a premium bakery chain based in the Netherlands. Established in 2018, the company has grown rapidly and now operates more than 30 branches across 20 major Dutch cities including Amsterdam, Rotterdam, Utrecht, The Hague, Eindhoven, and Groningen.

The bakery sells authentic Dutch baked goods from everyday bread like tijgerbrood and volkorenbrood, to traditional pastries like stroopwafels, appeltaart, bossche bollen, and seasonal specialties like oliebollen (only sold in December) and pepernoten (only sold during the Sinterklaas season in November).

The company employs over 450 people across all branches: bakers, assistant bakers, sales staff, branch managers, delivery drivers, and support staff. It serves tens of thousands of customers, many of whom participate in a loyalty program with tiered benefits (Bronze, Silver, Gold, Platinum).

The Current Problem

De Gouden Tarwe currently stores all its operational data in flat CSV files. As the business has grown, this approach has become unmanageable:

1. Data redundancy: The same product name, price, and description are repeated in every transaction record.
2. No formal relationships: There is no way to enforce that a branch referenced in an order actually exists.
3. No data integrity: A typo in a city name creates a new city that does not actually exist.
4. Impossible to answer business questions: Simple queries like What is our best-selling product in Rotterdam? require manually cross-referencing multiple files.
5. Performance problems: With millions of rows, opening and searching CSV files is painfully slow.

Progress – Day 1

On Day 1, we created the Entity Relationship Diagram from the CSV files. We determined the relations for each file to build the perfect database for this case. To determine the relations, data types, number of characters, and many more, we had to dive deep into the datasets. The ERD creation must follow database design principles, including normalization, naming conventions, soft deletes, and relationship rules.

Our datasets were split into 5 groups: geography and location, products and recipes, people, transactions, and also operations and supply chain. Geography and location contain regions, cities, branches, and branch operating hours. Products and recipes contain product categories, products, ingredients, product recipes, and price history. People contain customers, employees, and employee shift. The transactions group contains orders, order items, promotions, customer feedback, and loyalty transactions. Lastly, operations and supply chain

contain suppliers, purchase orders, purchase order items, inventory stock, daily production, waste log, and also equipment. Here is our ERD design.

